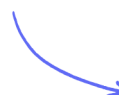


Structured Questions

Introduction to Kinetics & Equilibria

Collision Theory / Rates of Reaction / Maxwell-Boltzmann Distributions / Catalysts & Energy / Dynamic Equilibrium in Reversible Reactions / Le Chatelier's Principle / Industrial Compromises

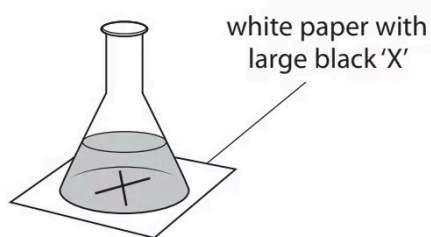
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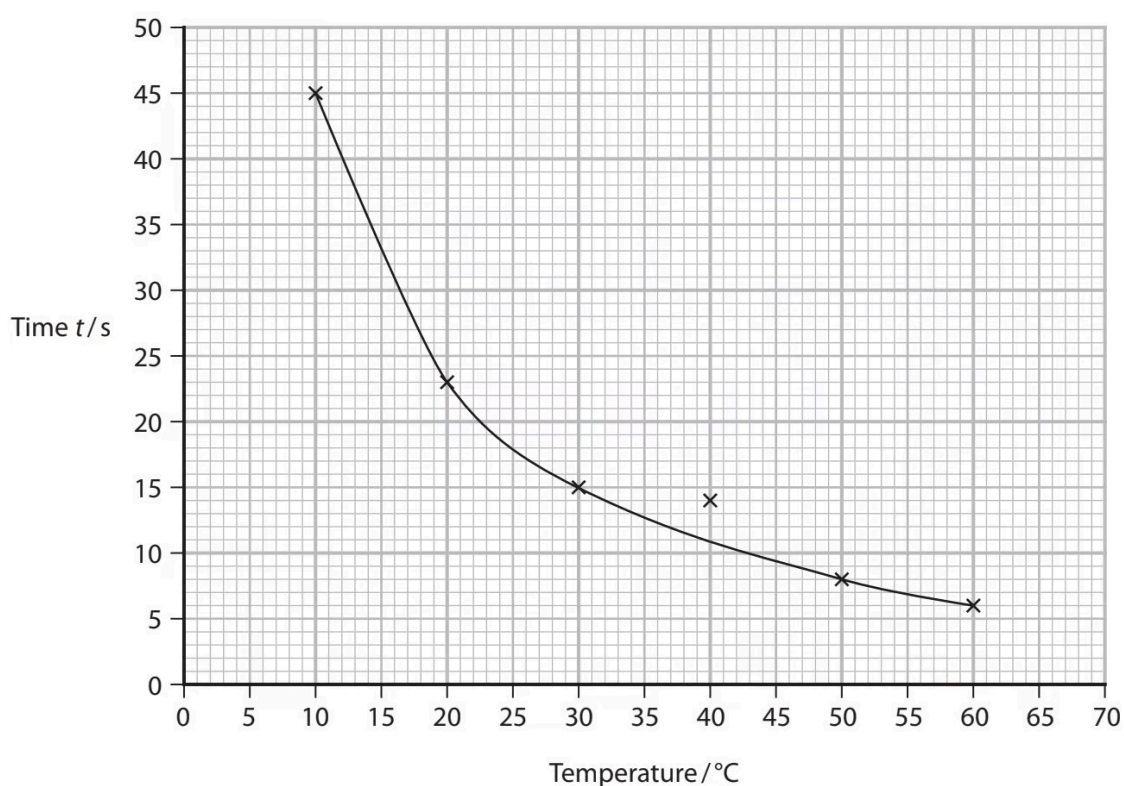
Total Marks

/58

- 1 (a) Students were set a challenge by their teacher to produce a chemical clock measuring a 20 s time interval. They used an opaque solution that became transparent, allowing a black cross to become visible after 20 s.



The students investigated the effect of temperature on their results and plotted a graph.



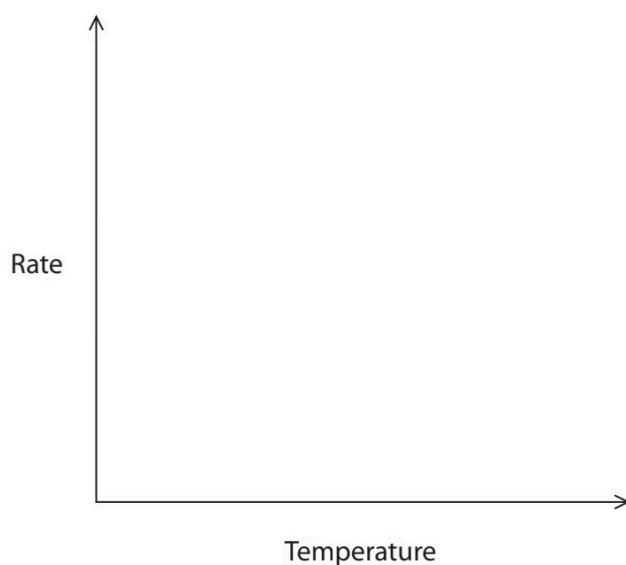
In this type of experiment $1/t$ (where t is time) may be used as a measure of the rate of reaction.

- i) Calculate the rate at 15°C to a suitable number of significant figures. Include units in your answer.

(3)

ii) Sketch a line showing how the rate of reaction varies with temperature for this reaction.

(1)



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(4 marks)

(b) Evaluate the students' results and decide whether it is necessary to repeat their experiments.

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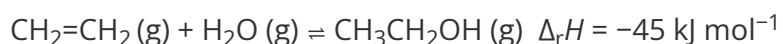
(2 marks)

(c) State how you would change the conditions to make this chemical clock measure 40 s at 22 °C.

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(1 mark)

- 2 (a)** Ethanol can be made in industry by the reaction of ethene with steam, using a phosphoric(V) acid catalyst.



The reaction is carried out at 300 °C and 60 atm. An initial yield of 5% is achieved when the ethene and steam first pass through the reactor.

Explain the chemical reasons for the conditions used and why such a low initial yield is acceptable in the industrial process.

(6 marks)

- (b)** The product of the reaction is a mixture of ethanol and water.

Explain why ethanol and water mix together fully.

You may find it helpful to draw a diagram.

(3 marks)

(c)]

Ethanol can be oxidised using a solution of acidified potassium dichromate(VI), $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$.

Ethanoic acid and another organic compound, **Y**, are both possible products.

i) Draw the structure of **Y**.

(1)

ii) State the conditions needed to maximise the yield of each product.

(2)

Conditions for maximum yield of **Y**

Conditions for maximum yield of ethanoic acid

(3 marks)

3 (a) This question is about ethanol and bioethanol.

The main fuel used as a petrol substitute is bioethanol. Bioethanol is ethanol that has been produced by fermentation. The starting material is usually some form of plant material rich in starch, such as wheat, maize or potatoes. Enzymes in yeast convert this material to simple carbohydrates such as glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and then to ethanol and carbon dioxide.



The mixture is left for several days until fermentation is complete. The percentage of ethanol is never greater than 15% because higher concentrations of ethanol kill the yeast.

A common blend of fuel is 95% petrol and 5% bioethanol. The engine does not need to be modified for this mixture.

Give **one** advantage and **one** disadvantage of using bioethanol in petrol.

Advantage

Disadvantage

.....
.....
(2 marks)

(b) Suggest why this fermentation must be carried out in the absence of air.

.....
(1 mark)

(c) Suggest how the ethanol can be obtained, after filtering the fermentation mixture.

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(1 mark)

(d) Ethanol is hygroscopic, which means it readily absorbs water from the air.

i) Give a possible reason why ethanol is able to absorb water.

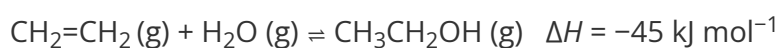
(1)

ii) Suggest a problem arising from the hygroscopic nature of ethanol when using this fuel in a motor vehicle.

(1)

(2 marks)

(e) Ethanol can also be produced by the hydration of ethene.



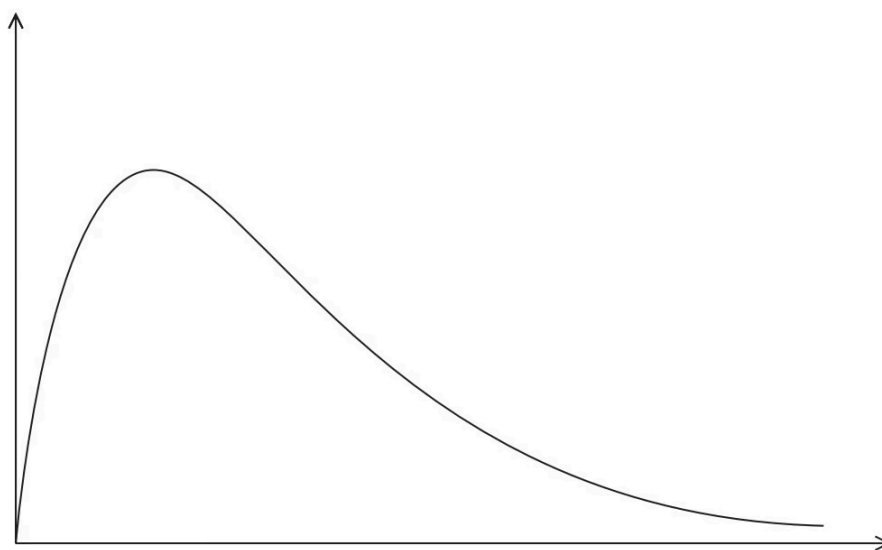
i) Typical conditions are 300 °C and 60 atm with a catalyst of phosphoric acid.

Explain why these conditions are used, by describing the effect of changing the temperature and pressure on rate of reaction, equilibrium yield and cost.

(6)

ii) The rate of this reaction is increased by using a catalyst of phosphoric acid. Label the axes on the Maxwell-Boltzmann distribution curve and use it to explain how a catalyst increases the rate of reaction.

(4)



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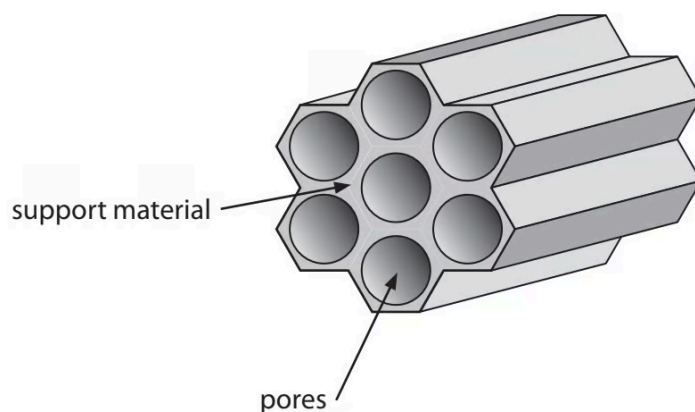
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(10 marks)

(f) Catalysts such as phosphoric acid are bonded to a support material that contains lots of

pores.



i) Suggest the advantage of using support materials containing lots of pores.

(1)

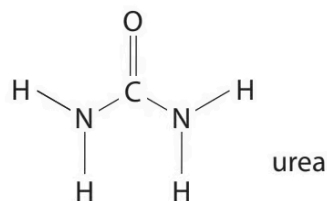
ii) Under these conditions, only about 5% of the ethene is converted into ethanol as it passes over the catalyst.

Suggest how the overall yield of this process can be improved to make it economically viable.

(2)

(3 marks)

- 4 (a)** Some diesel cars contain an extra catalytic converter for the reduction of nitrogen oxides (NO_x) in exhaust gases. A solution of urea is used for this process.



Urea has a melting temperature of 133°C .

Explain why this value is higher than expected for a relatively small molecule.

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(3 marks)

- (b)** A saturated solution of urea has a concentration of 9.07 mol dm^{-3} at 25°C .

Calculate the mass of urea in 150 cm^3 of a saturated solution.

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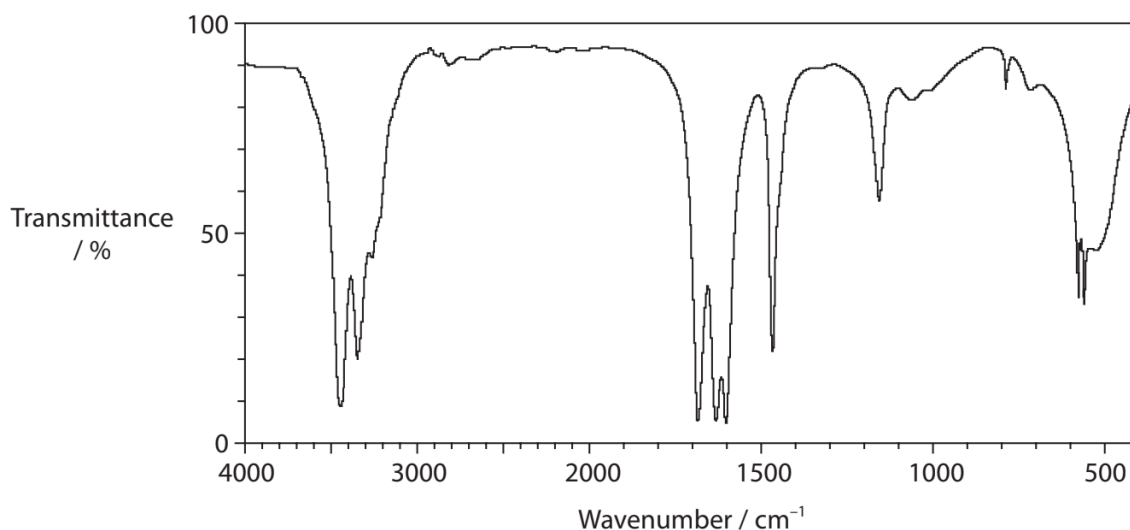
(2 marks)

- (c)** State why NO_x emissions are harmful to the environment.

.....

(1 mark)

(d) An infrared spectrum of urea is shown. Refer to your Data Booklet.



i) Draw a circle around an absorption in the spectrum that could be due to the stretching of the N—H bond.

(1)

ii) Identify the bond responsible for the absorption at 1683 cm^{-1} .

(1)

(2 marks)

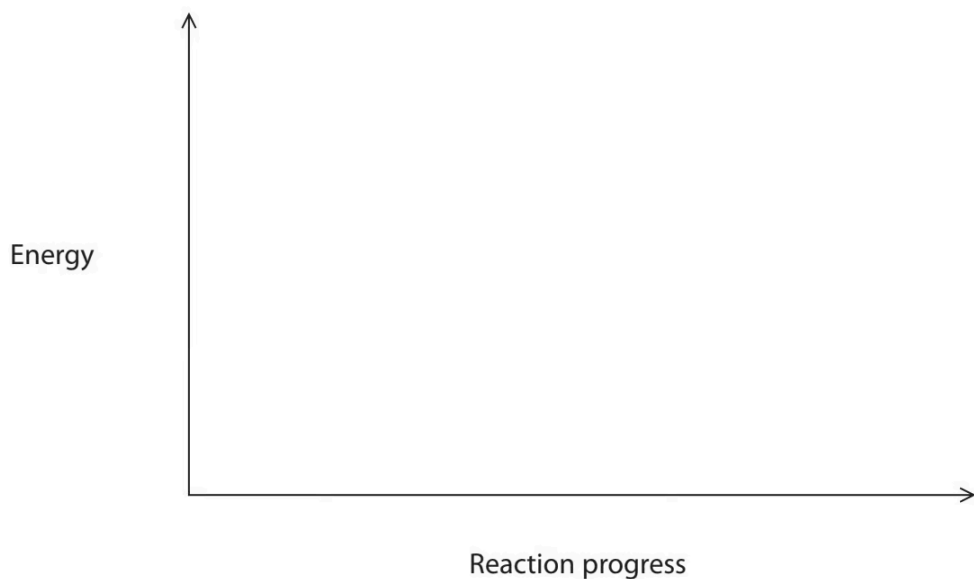
(e) In a diesel car exhaust system, the urea reacts with water to form ammonia and carbon dioxide. The enthalpy change for this reaction is $+133\text{ kJ mol}^{-1}$.

i) Complete the equation for this **reversible** reaction. State symbols are not required.

(1)



ii) Sketch the reaction profile for the forward reaction on the axes provided. Include labels for ΔH and the activation energy (E_a).



(3)

(4 marks)

(f) The catalytic converter contains metal oxides. When the exhaust gases pass through the catalytic converter, ammonia reacts with NO_x gases to form nitrogen and water.

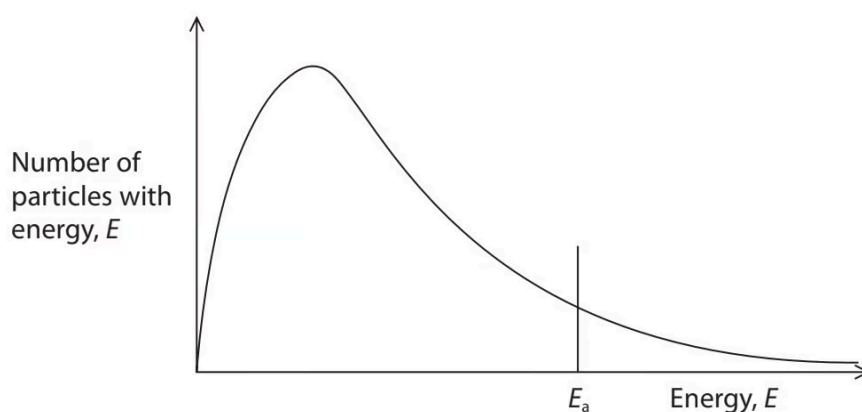
i) Explain why it is **not** correct to state that urea is acting as a catalyst in the reaction.

(1)

ii) Explain how a catalyst increases the rate of a chemical reaction.

Use the Maxwell-Boltzmann distribution shown and refer to the collision theory.

(3)



(4 marks)

(g) The catalytic converter works best at a temperature of around 350 °C.

i) Suggest how the catalytic converter reaches this temperature.

(1)

ii) The chemical reactions in the exhaust system of a diesel car, using a catalytic converter, form 89.3 m³ of nitrogen per hour.

Calculate the number of molecules of nitrogen formed per hour.

[Molar volume at 350 °C = 51.1 dm³ mol⁻¹ Avogadro constant, $L = 6.02 \times 10^{23}$ mol⁻¹]

(3)

(4 marks)